Modeling the Effect of Pitch Sequencing and Batter-Specific Adjustments on Swing-and-Miss Outcomes in MLB

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1 Introduction

The rise of advanced tracking systems in all sports, especially Major League Baseball (MLB) has transformed how teams evaluate pitching performance and decide on their pitching strategy. Since the introduction of Statcast in 2015, every pitch thrown in any MLB game has been measured through many different metrics such as release speed, spin rate, movement profile, release point, and extension. All these metrics are captured at the millisecond (Major League Baseball, 2024). This sudden increase in analytics has led to massive changes in pitcher development, with front offices investing heavily in how pitchers are drafted and managed throughout their tenure. All of this is done to maximize the effectiveness of each pitch in a pitcher's arsenal. This leads to the central question in baseball analytics: *what attributes make a pitch difficult to hit?* Intuitively, batters and most baseball watchers agree that pitches with higher velocity, greater movement, and longer extension are harder to hit. However, standard logistic regression models that predict swing-and-miss probability based solely on these physical characteristics often lead to inconclusive results. In such models, the probability of a swing-and-miss, $p_i = P(Y_i = 1)$, is typically described as:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1\,\texttt{Speed}_i + \beta_2\,\texttt{IVB}_i + \beta_3\,\texttt{HB}_i + \beta_4\,\texttt{Ext}_i + \beta_5\,\texttt{Arm}_i \tag{1}$$

where `IVB` denotes induced vertical break, `HB` horizontal break, `Ext` release extension, and `Arm` arm angle. Previous such studies have also found that main effects and simple interactions in this framework are often statistically insignificant or fail to explain the variance in outcomes. The limitation of this approach is that it treats each pitch as an independent, context-free event, ignoring two critical sources of variation: **sequencing** and **batter heterogeneity**. First, pitches are part of a strategic sequence. A pitch's effectiveness cannot be explained solely based on its individual metrics, but rather the buildup to that pitch. For example, a breaking ball may induce more whiffs when it follows a fastball due to the disruption in batter timing and visual processing, a phenomenon also known as pitch tunneling. Second, batters differ significantly in their baseline contact rates. Elite contact hitters like Shohei Ohtani or Aaron Judge rarely swing and miss regardless of pitch quality, while high-strikeout batters may whiff at average pitches. Neglecting these batter-specific tendencies introduces a lot of noise into the analysis of this dataset, which in turn hinders the conclusions that we can draw about the true relationship between pitch quality and outcomes. To address these limitations, we want to propose a refined framework that tackles this issue by incorporating pitch sequencing and batter heterogeneity. Specifically, our research question is:

How do pitch sequencing (the type, location, and result of the previous pitch) and batter-specific tendencies jointly influence the probability of a swing-and-miss, beyond the physical characteristics of the pitch itself?

To investigate this, we propose the following objectives:

1. **Pitch Sequence Effects:** Investigate whether the previous pitch type and result significantly affect the whiff probability of the current pitch. We believe that specific sequence combinations (e.g., Fastball → Slider) will show significant interaction effects.
2. **Differentiate between Batter skill and Pitch quality:** Implement a Generalized Linear Mixed Model (GLMM) where individual batters are treated as random effects. This will help separate the variance attributable to the pitch from the variance attributable to the batter's skill.
3. **Prediction Power:** Evaluate whether the Sequencing and Mixed-Effects models provide a better fit and have more predictive power compared to the baseline metrics that a single pitch provides.

2 Research Strategy

2.1 Data

We will utilize pitch-by-pitch data from the 2025 Major League Baseball (MLB) season obtained from the Statcast database via the `pybaseball` Python module. Statcast provides high-resolution tracking data for every pitch thrown in MLB games. Our unit of observation is an individual pitch.

For each pitch i, we define a binary outcome:

$$Y_i = \begin{cases} 1 & \text{if the batter swings and misses (whiff),} \\ 0 & \text{otherwise.} \end{cases}$$

Thus, $p_i = P(Y_i = 1)$ represents the probability of a swing-and-miss on pitch i.

We group predictors into three separate categories:

(1) Physical Pitch Characteristics

- `Speed` (→ `release_speed`): Release velocity (mph)
- `IVB` (→ `pfx_z`): Induced vertical break

- `HB` (→ `pfx_x`): Horizontal break
- `Ext` (→ `release_extension`): Release extension (feet)
- `Arm`: Arm angle at release

(2) Sequencing Variables

- Previous pitch type
- Previous pitch location
- Previous pitch outcome
- Interactions between current and previous pitch types

(3) Batter-Specific Effects

To account for heterogeneity in contact ability and baseline swing-and-miss tendencies, batter identity is incorporated as a random effect within a generalized linear mixed model (GLMM).

In practice, we define a whiff using Statcast's `description` values (e.g., `swinging_strike`, `swinging_strike_blocked`). Lagged sequencing variables are constructed within the same plate appearance (first-pitch lags are treated as missing). We restrict to regular-season pitches with complete measurements and exclude obvious measurement errors and non-competitive events (e.g., intentional walks).

2.1.1 Data Cleaning and Exploratory Analysis

We perform rigorous data cleaning to ensure the reliability of our measurements. Pitches with unrealistic physical values, such as release speeds below 65 mph or undefined movement metrics, are treated as tracking errors and removed. We also filter out non-competitive events, including pitchouts and intentional walks, as these do not reflect genuine pitcher-batter interactions.

To facilitate the sequencing model, we construct lag variables for the previous pitch's type, location, and result. These variables are generated strictly within specific plate appearances; for the first pitch of an at-bat, the previous pitch characteristics are coded as a distinct "None" category to preserve the observation.

Variable	Mean	SD	Min	Max
Speed (mph)	93.4	3.2	68.5	100.1
IVB (in)	14.1	5.8	-5.2	22.4
HB (in)	6.5	4.1	-10.1	20.3

Table 1: Summary statistics for key physical pitch characteristics in the cleaned 2025 dataset.

2.2 Planned Analysis

We will model the binary outcome, p_i, the probability of a swing-and-miss on pitch i, thus our models will be estimated using logistic regression. A standard model predicting p_i based solely on physical pitch characteristics (vector X_i), identical to equation (1), serves as the nested baseline for our our models:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1\texttt{Speed}_i + \beta_2\texttt{IVB}_i + \beta_3\texttt{HB}_i + \beta_4\texttt{Ext}_i + \beta_5\texttt{Arm}_i \\ = \text{logit}(p_i) = \beta X_i \tag{2}$$

To address the inconclusive results seen from models purely considering physical characteristics, our first complex model will modify equation (2) to account for the strategy of pitch sequencing common at the highest levels of baseball. We will introduce lagged variables (vector Z_i) on the previous pitch type, location, and outcome to capture sequencing, and use interaction effects between current and previous pitch types (vector $T_i * T_{i-1}$) to find the effect of different pitch type sequences. Thus, we estimate this logit Sequencing Model as follows:

$$\text{Sequencing Model: } \text{logit}(p_i) = \beta X_i + \gamma Z_{i-1} + \alpha(T_i * T_{i-1}) \tag{3}$$

Our second complex model incorporates mixed effects to separate the batter's skill from the pitch itself, hence why we will assign batters as random effects. As previously, it is an expanded model of (2), with the physical pitch characteristics serving as the fixed effects as before. The batter specific random effects (scalar b_j) are random intercepts indexed by batter j, with each b_j assumed to be normally distributed with mean 0 and variance σ^2.

$$\text{GLMM: } \text{logit}(p_i) = \beta X_i + b_j \text{ where } b_j \sim \mathcal{N}(0, \sigma^2) \tag{4}$$

To determine if our added parameters accounting for pitch sequencing, interaction effects, and batter heterogeneity significantly improve the fit to the data, we will use Likelihood Ratio Tests (LRTs). We will perform an LRT for comparison of the physical pitch characteristics model (2) with the sequencing model (3), and for the physical pitch characteristics model (2) with the GLMM for batter heterogeneity (4). Then, BIC will be employed to identify the best fitting and most parsimonious of the three models, as (3) and (4) are not nested.

Avoiding model misspesification is critical to the validity of our models, and generating meaningful results. To monitor diagnostics, Variance Inflation Factors (VIFs) will be computed to evaluate possible multicollinearity between predictors, and residual plots will assess the linearity of predictors assumed in each model. To check the normal distribution assumption of the random effects in the GLMM, a Q-Q plot will be used. Other diagnostics will also be explored to test for autocorrelation in the Sequencing Model and overdispersion in the GLMM.

References

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